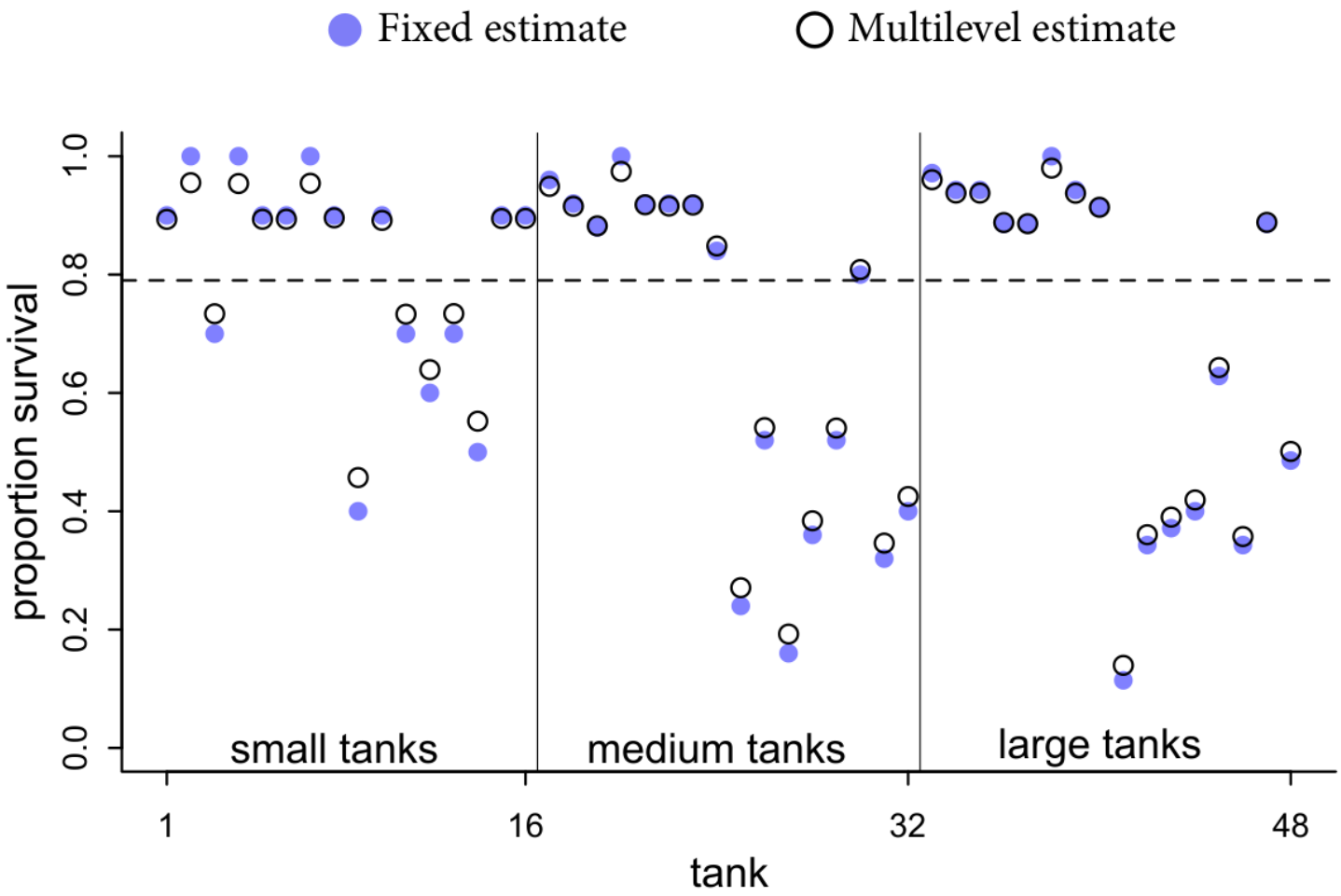
Lecture 36: Multi-level models - continued

Tadpoles example from the previous lecture



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UNIVERSITY OF NEW ZEALAND

Simulating tadpoles

Principles

- To demonstrate the complete, no and partial pooling effects, we will simulate tadpole data.
- Simulations: you know the true per-pond survival probabilities. And potenttially relationship to other variables (density, predator presence...)
- Can compare them to estimated values
- **Learning to simulate and validate models and model fitting is very valuable**
- You simulate data with known specified parameter values and then you make sure your estimation method can recover these values with a tolerable range of precision
- Even to understand the model only.

Simulating tadpoles in **ponds** - The model

$$
\begin{aligned}
S_i &\sim \text{Binomial}(N_i, p_i) \\
\text{logit}(p_i) &= \alpha_{Pond[i]} \\
\alpha_j &\sim \text{Normal}(\overline{\alpha}, \sigma) \\
\overline{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

Need to assign (in order) values to:

1. $\overline{\alpha}$ (average log-odds of survival in the entire pond population) and σ (sd of the log-odds of survival in the entire pond population)
2. α a vector of individual pond intercepts (one for each pond)
3. Last, we SIMULATE S_i's, the number of survival tadpoles, each of them assigned to one pond.

Note that priors are part of the estimation model, not the simulation model. They are epistemiological, not ontological. The represent the initial state of knowledge of our engine, not a statement about how nature chooses parameter values.

Tadpole simulation

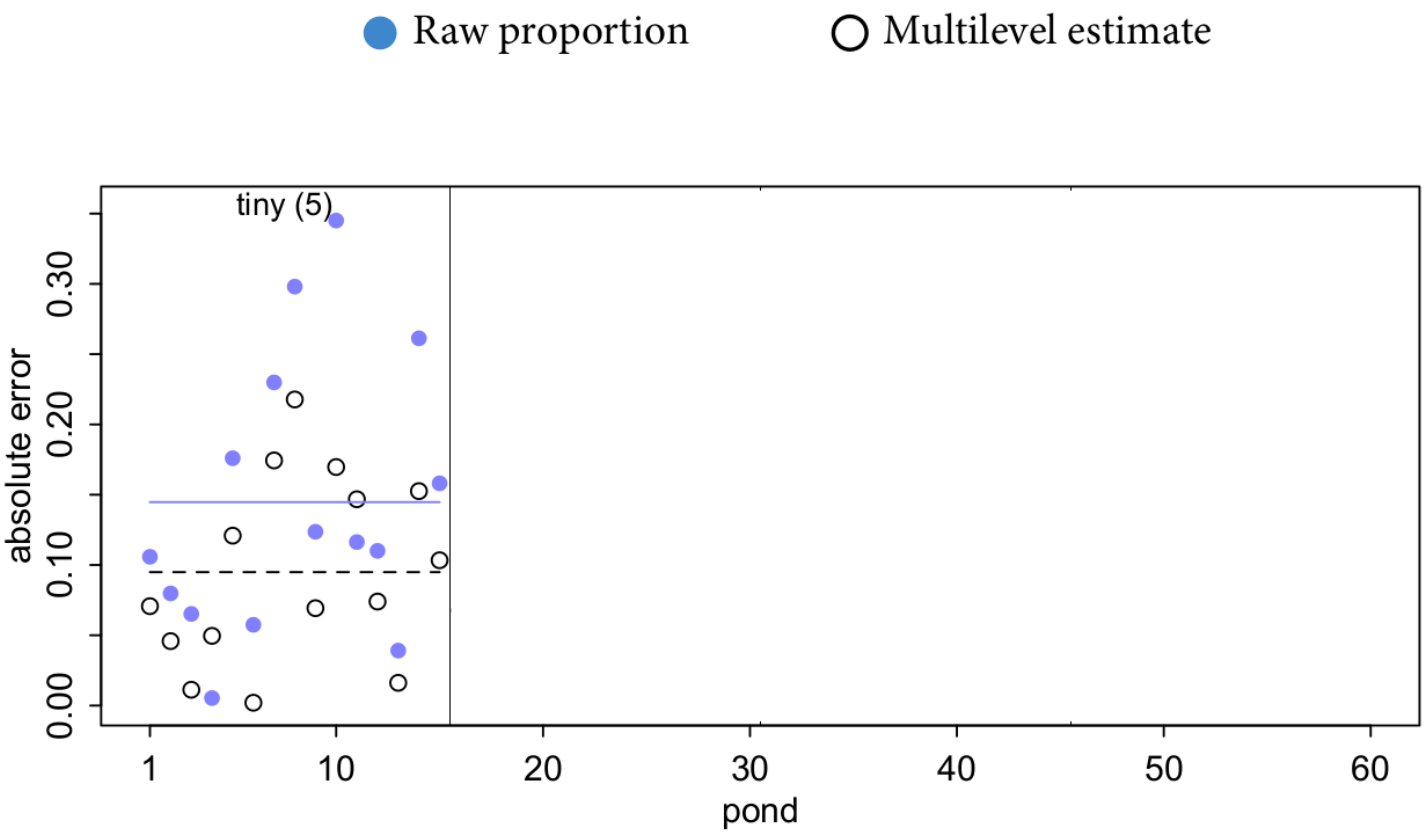
The data

- $n_{ponds} = 60$ ponds
- 5, 10, 25 and 35 tadpoles, each replicated $N_i = 15$ times.
- $\overline{\alpha} = 1.5$, $\sigma = 1.5$
- See the simulation code in Section 13.2 of the SR book

	pond	n	true.a	s	p.nopool	p.partpool	p.true
1	1	5	-3.089936132	1	0.2000000	0.32173203	0.04352429
2	2	5	0.267290817	5	1.0000000	0.91305884	0.56642768
3	3	5	0.896554101	4	0.8000000	0.79164823	0.71024085
4	4	5	1.934806220	5	1.0000000	0.91276066	0.87378044
5	5	5	-0.758682067	0	0.0000000	0.17692527	0.31893247
6	6	5	3.904836388	5	1.0000000	0.91337140	0.98025353
7	7	5	2.271914139	4	0.8000000	0.79349508	0.90652411
8	8	5	2.886101619	4	0.8000000	0.79557800	0.94715510
9	9	5	1.436457877	3	0.6000000	0.64219989	0.80790553
10	10	5	1.156079068	3	0.6000000	0.64414477	0.76061953

Model estimation output

|Error| in average pond survival probability

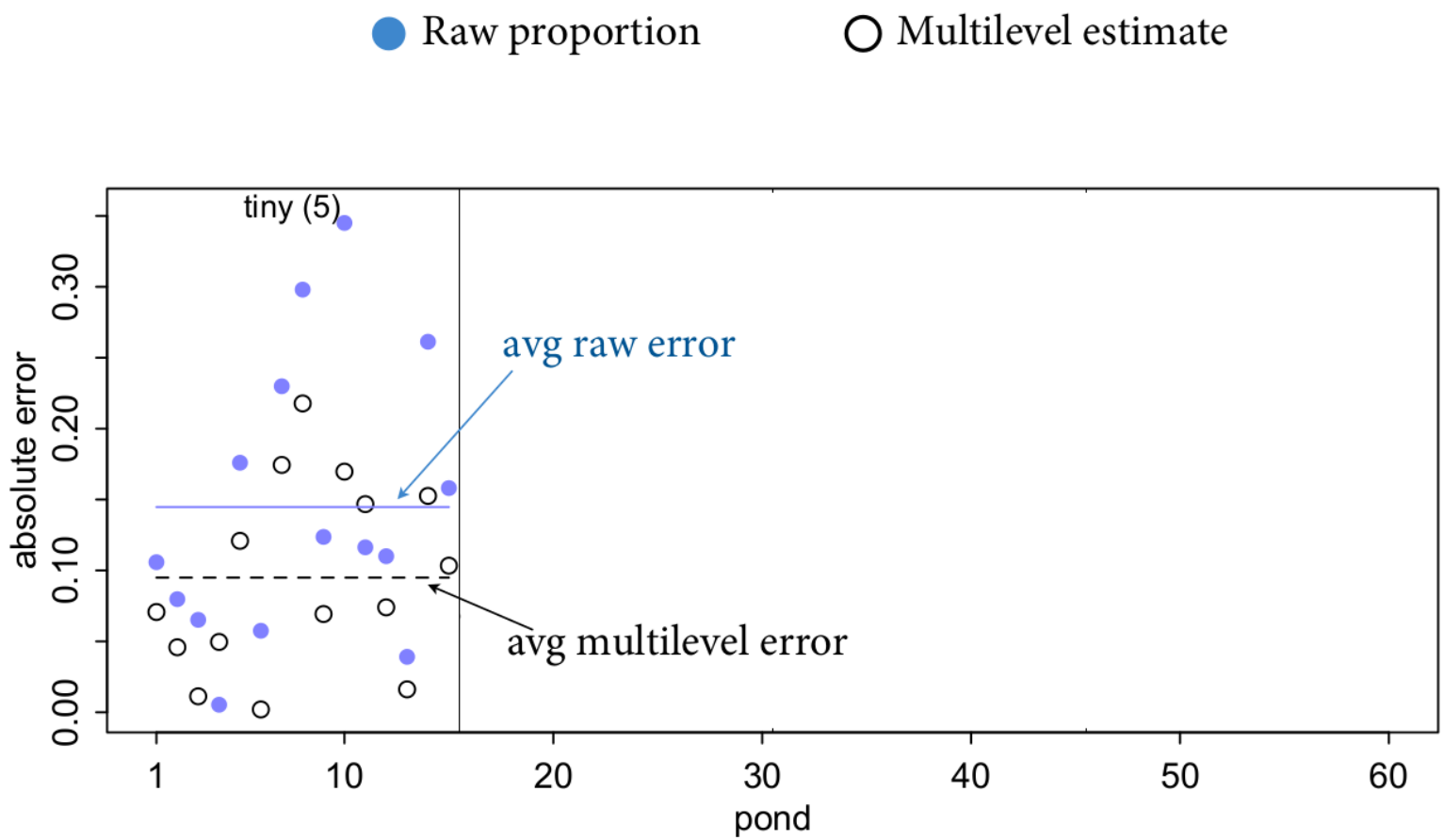


The closer to the bottom, the better.

Small sample size in small ponds makes inference hard. But less hard for pooled estimation.

Model estimation output

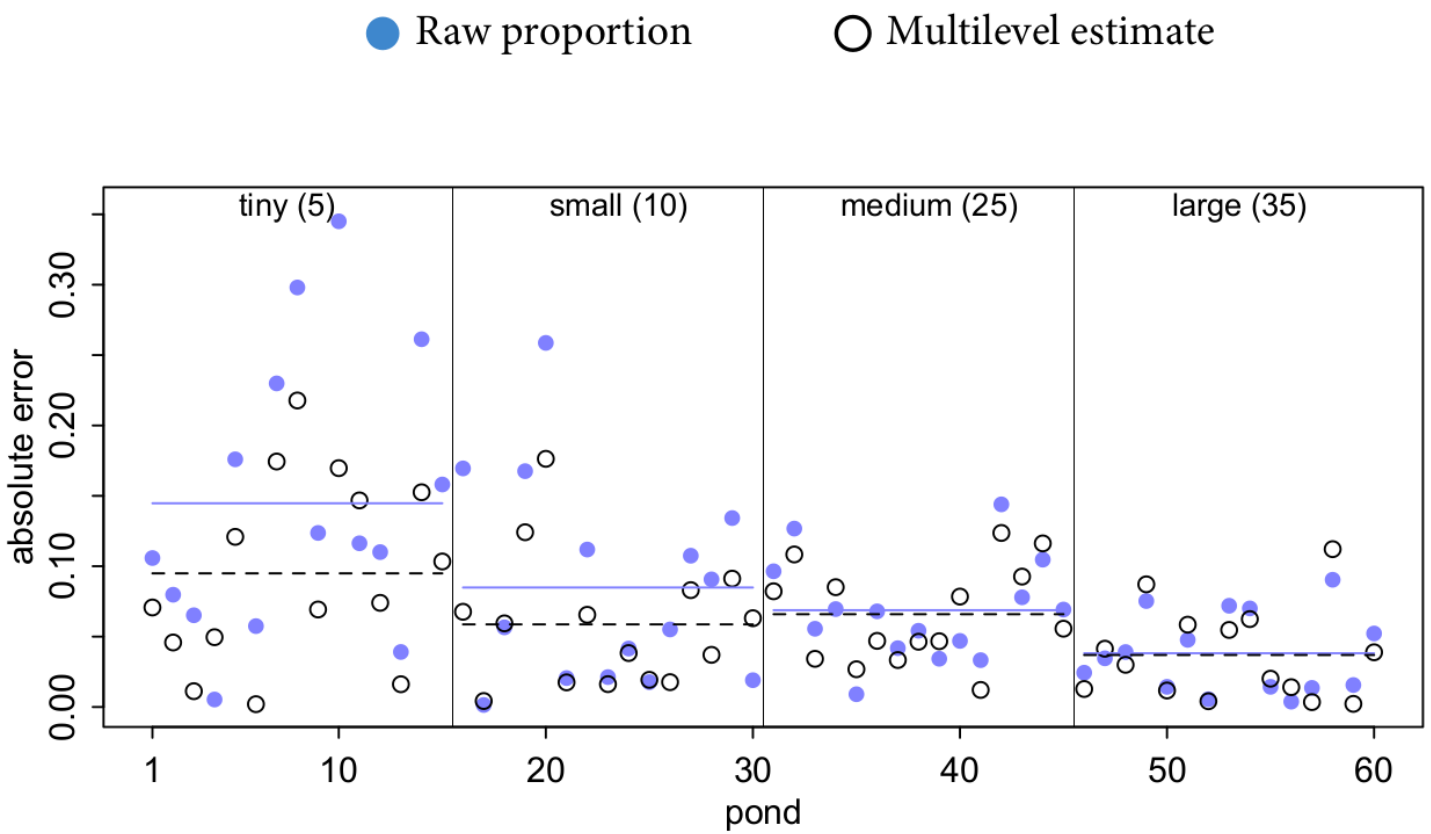
|Error| in average pond survival probability



Should repeat this scenario to see trend, we have simulation randomness, so cannot conclude.

Model estimation output

|Error| in average pond survival probability

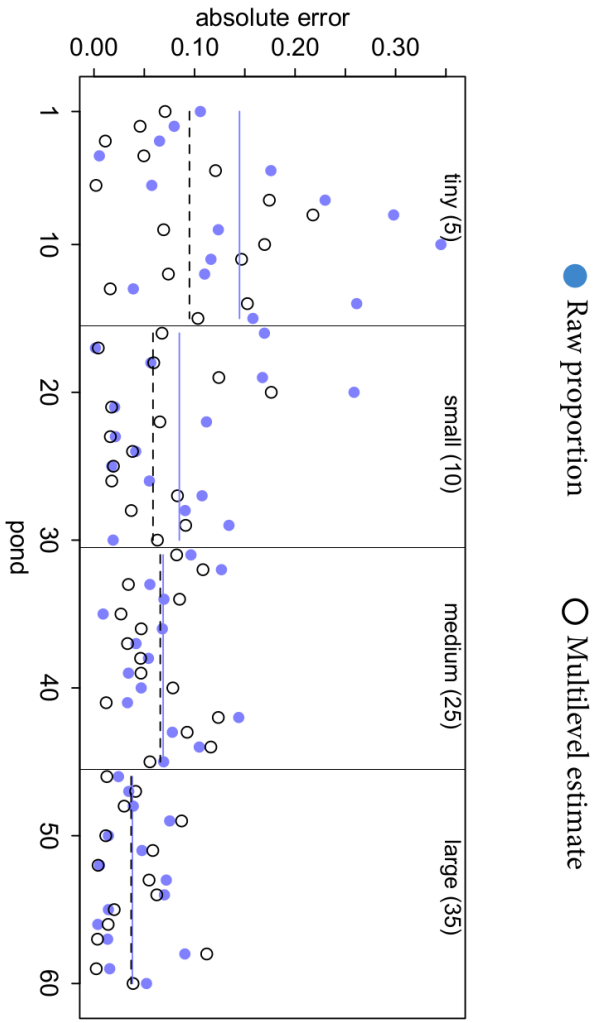


Larger ponds lead to more accurate estimation, whatever the method (no confounding).

Blue line always above the black dashed line (but medium ponds).

Shrinkage

- Varying effect estimates *shrink* towards mean $\overline{\alpha}$.
- Further from the mean, more shrinkage
- Fewer data in cluster, more shrinkage
- Same as regression to the mean, really
- Shrinkage results from pooling of information
- Multi-level models reduce fit to sample...but that's better because they give better estimates of the underlying process!



Beyond varying intercept

- More varying intercept with more than one type of clusters (Section 13.3). Means different posteriors for different types of variations (Section 13.5)
- Copying with divergent transition: centred vs non-centred parametrisations (Section 13.4)
- Varying slopes (Section 14.1)
- Instrumental variables (natural experiment on the exposure) and causal designs: go multivariate (Section 14.3)
- Measurement error (Section 15.1) and Missing data (Section 15.2)
- More complex models (e.g. see the Prey-Predator model in Section 16.4)

High-standards in science

- Quality of theory and predictions. A good theory leads precise predictions that can be tested. Use more than one model, challenge them.
- Money...who receive funding for their work? Long-term funding is crucial, not (only) short-term benefits! Also conservative (and corrupt) if lack of diversity.
- Quality of measurements and research design. Low signal-to-noise ratios will lead to lowly reliable findings.
- Quality of data analysis: design an adequate analysis. Don't please the data. Or your a priori.
- Publication: (good) prepublication peer-review is invaluable. Not perfect. Careful with incentive for positive findings that distort research. Watch for post-publication comments and retracted studies (still get citations!?!)
- Replication and meta-analysis: repeat, repeat, repeat...and synthetise. Journals prioritise "new" findings. If the literature is biased, aggregating it magnifies the bias :-/.

The end

Thank you!

It's been a pleasure to teach you this course.

Please take the time to answer this survey https://www.surveymonkey.com/r/9BM793K anonymously or feel free to sign it

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The exam

General

- All questions in one pdf file coined `exam161304_S2_2022_final.pdf`
- Three questions worth 28, 43 and 38 marks, respectively
- It is long, no panic, I know
- The goal is to get you thinking and to show me your reasoning. You might get bonus point if you show pertinent thinking to questions instead of rushing to try and fish marks

The exam

R

- Some questions will use R, indicated by the  logo.
- A companion `exam161.304-S2-2022-Rscripts.R` file contains all the code shown in the exam
- You can run all the code you want, only three (3) items require you to run some code (may need to write some bits) and show/comment on the outputs.
- If your code does not run, explain what you were trying to do and the logic of it, it might grant you (almost) full marks.

The exam

Topics

- Linear regressions, simple and multiple, writing models, interpreting coefficients and outputs
- With factor and continuous predictors
- Thinking about causal effects and bias
- DAGs
- Prior checks and posterior plots (of parameter distribution, prediction, contrasts)
- Model comparison
- Over- and under-fitting considerations (one item on multi-level models only)
- Logistic regression
- Simulations for model checking